AI2: Bayesian Networks – Learning and Inference

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Outline

We assume throughout that the dependency structure of the Bayes net is known.

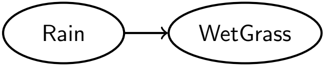
Learning the conditional probabilities from data is most commonly done by MLE (counting). MAP can also be used instead.

In this lecture we learn:

- Exact Inference
  - A worked example: "Wet Grass"

Example of determining parameters

Task: Construct the CPTs for this graph.



Suppose know (from prior knowledge, or estimated from some data):

- The probability of raining through a day is:

$$P(R = 1) = 0.4 \Rightarrow P(R = 0) = 0.6$$

- The probability that the grass gets wet when it rains is:

$$P(W = 1 \mid R = 1) = 0.9 \Rightarrow P(W = 0 \mid R = 1) = 0.1$$

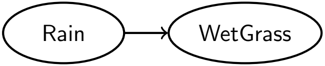
- The probability that the grass gets wet without raining (e.g., sprinkler on) is:

$$P(W = 1 \mid R = 0) = 0.2 \Rightarrow P(W = 0 \mid R = 0) = 0.8$$

Example of determining parameters

Solution: The relevant CPTs are:

R	$P(R)$
0	0.6
1	0.4



R	$P(W = 0 \mid R)$	$P(W = 1 \mid R)$
0	0.8	0.2
1	0.1	0.9

What is Inference in Bayesian Networks?

Inference in Bayesian networks answers 2 types of questions:

- Diagnosis: $P(Cause|Effect)$, e.g. $P(R|W)$
- Prediction: $P(Effect|Cause)$, e.g. $P(W|C)$

Applications of inference in Bayesian networks:

- Classification: $\max_{\text{class}} P(\text{class}|\text{data})$
- Decision Making: $\sum_i P(Effect_i|Cause) \times Utility(Effect_i|Cause)$

Example of making inference (in the simple direct influence network)

In the "direct influence" structure, there are 2 random variables (nodes). **Inference** is the problem of finding out the "cause" variable when we only observe the "effect" variable.

- **Observed Variables:** The ones we have knowledge about.
- **Unobserved (hidden) Variables:** The ones we do not observe.

Question: If you observe the grass is wet, what's the probability it rained?
Solution: By Bayes' rule,

$$P(R = 1 \mid W = 1) = \frac{P(R = 1)P(W = 1 \mid R = 1)}{P(W = 1)}$$

Calculate the **marginal probability** of Wet Grass:

$$P(W = 1) = \sum_{r \in \{0,1\}} P(W = 1, R = r) = \sum_{r \in \{0,1\}} P(W = 1 | R = r)P(R = r)$$
$$= (0.9 \times 0.4) + (0.2 \times 0.6) = 0.48.$$

Example of making inference (cont'd)

Plugging this back into the denominator of Bayes' rule,

$$P(R=1 \mid W=1) = \frac{P(R=1)P(W=1 \mid R=1)}{P(W=1)} = \frac{0.9 \times 0.4}{0.48} = 0.75$$

Interpretation:

- The above inference is called **diagnosis**, i.e., to obtain $P(\text{Cause} \mid \text{Effect})$.
- **Conclusion:** Knowing that the grass is wet increased the probability it rained from $P(R=1) = 0.4$ to $P(R=1 \mid W=1) = 0.75$.

Inference in more general: Three sets of variables

For inference purposes, we distinguish 3 sets of variables (nodes) in a Bayes net:

- Evidence variables $E = \{E_1, \ldots, E_m\}$ (known)
- Query variables $X = \{X_1, \ldots, X_n\}$ (wanted)
- Non-evidence variables $Y = \{Y_1, \ldots, Y_k\}$ (neither known nor wanted, but must deal with)

The complete set of variables (nodes) of a Bayesian network is the union of these: $E \cup X \cup Y$.

Inference in Bayesian Networks

Inference (aka. Query): Answering questions about the underlying probability distribution.

- **Unconditional or conditional probability inference:**
  - What is the probability of a given value assignment for a subset of variables in X? Example: $P(W = 1)$
  - What is the probability of different value assignments for query variables X given evidence about variables E: Compute $P(X|E = e)$. Example: $P(W|C = 1)$
- **Maximum a posteriori (MAP) inference:**
  - Given evidence $E = e$, find the most likely assignment of values to the query variables X:

    $$\mathrm{MAP}(X|E = e) = \arg\max_{x} P(X = x|E = e)$$

    Example: Classification. (Given training data and a new example, determine the most probable class label.)

Exact Inference in Bayesian Networks

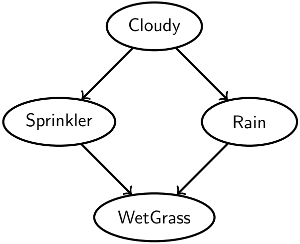
Exact inference computes the posterior probability distribution. The general procedure is as follows:

$$P(X|E) = \frac{P(X,E)}{P(E)} = \alpha \cdot P(X,E) = \alpha \cdot \underbrace{\sum_{Y} P(X,E,Y)}_{Marginalisation} \qquad (1)$$

where $\alpha = \frac{1}{P(E)}$.

Bayesian Networks: Wet Grass Example

C	$P(C)$
0	0.4
1	0.6



C	$P(S=0 \mid C)$	$P(S=1 \mid C)$
0	0.5	0.5
1	0.9	0.1

C	$P(R=0 \mid C)$	$P(R=1 \mid C)$
0	0.8	0.2
1	0.2	0.8

S	R	$P(W=0 \mid S,R)$	$P(W=1 \mid S,R)$
0	0	1.0	0.0
1	0	0.1	0.9
0	1	0.1	0.9
1	1	0.01	0.99

Bayesian Networks: Wet Grass Example

In a Bayesian network, we always have have:

$$P(X_1, X_2, \ldots, X_n) = \prod_{i=1}^{n} P(X_i|\text{Parents}(X_i))$$

Expanding for the Wet Grass example:

$$P(C, S, R, W) = P(C)P(S|C)P(R|C)P(W|S, R)$$

where the nodes are: Cloudy (C), Sprinkler (S), Rain (R), Wet Grass (W).

Question: Given that it is cloudy ($C = 1$), what is the probability that the grass is wet ($W = 1$)?

Worked Example: "Wet grass" (part 1)

$$P(C, S, R, W) = P(C)P(S|C)P(R|C)P(W|S, R)$$

Question: Given that it is cloudy ($C = 1$), what is the probability that the grass is wet ($W = 1$)?

Worked Example: "Wet grass" (part 1)

$$P(C,S,R,W) = P(C)P(S|C)P(R|C)P(W|S,R)$$

Question: Given that it is cloudy ($C = 1$), what is the probability that the grass is wet ($W = 1$)?

Solution: The required probability can be computed as:

$$P(W = 1|C = 1) = \frac{P(W = 1, C = 1)}{P(C = 1)}$$

where $P(C = 1) = 0.6$, and we compute the numerator by marginalizing over S and R:

$$P(W = 1, C = 1) = \sum_{S \in R_S} \sum_{R \in R_R} P(W = 1, S, R, C = 1)$$

$$= \sum_{S} \sum_{R} P(C = 1)P(S|C = 1)P(R|C = 1)P(W = 1|S, R)$$

More on Marginalization

Marginal distribution: A distribution formed by calculating the subset of a larger probability distribution of a collection of random variables.

Example: Two random variables X and Y with joint PMF and marginal PMFs:

	$Y = 0$	$Y = 1$	$Y = 2$	$P_X(x)$
$X = 0$	$4/24$	$6/24$	$3/24$	$13/24$
$X = 1$	$3/24$	$4/24$	$4/24$	$11/24$
$P_Y(y)$	$7/24$	$10/24$	$7/24$	$24/24$

Calculation:

- For discrete random variables, we **sum** over the unwanted variable.
- For continuous random variables, we would **integrate** over the unwanted variable.

More on Marginalization

Calculation: For continuous random variables, we would **integrate** over the unwanted variable.

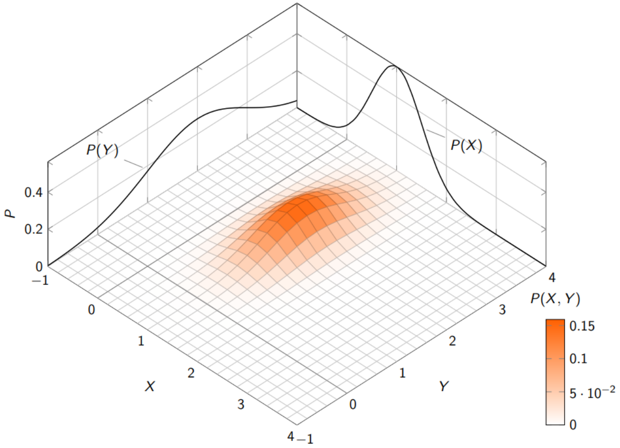


Figure: Example PDF of a bivariate normal distribution $p(X, Y)$, and its marginal distributions $P(X)$ and $P(Y)$. Source: Stackexchange: Tex

Worked Example: "Wet grass" (part 2)

$$P(W=1|C=1) = \frac{\sum_S \sum_R P(W=1, S, R, C=1)}{P(C=1)}$$

$$\frac{\sum_S \sum_R P(W=1|S,R)P(S|C=1)P(R|C=1)P(C=1)}{P(C=1)}$$

$$= \ldots$$

C	$P(C)$
0	0.4
1	0.6

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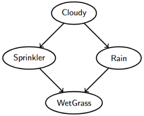

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0	0.8	0.2
1	0.2	0.8

S	R	$P(W=0\|S,R)$	$P(W=1\|S,R)$
0	0	1.0	0.0
1	0	0.1	0.9
0	1	0.1	0.9
1	1	0.01	0.99

Worked Example: "Wet grass" (part 2)

$$P(W=1|C=1) = \frac{\sum_S \sum_R P(W=1,S,R,C=1)}{P(C=1)}$$

$$\frac{\sum_S \sum_R P(W=1|S,R)P(S|C=1)P(R|C=1)P(C=1)}{P(C=1)}$$

$= \ldots$

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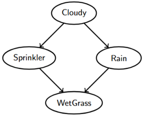

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0	1	0.1	0.9
1	1	0.01	0.99

Worked Example: "Wet grass" (part 2) - solution

Observe that $P(C = 1)$ cancels out (from both numerator and denominator), and from the tables given earlier we have (do it yourself!):

$$P(W = 1|C = 1) = \sum_{S \in R_S} \sum_{R \in R_R} P(W = 1|S, R) P(S|C = 1) P(R|C = 1) = \ldots$$

C	$P(S = 0\|C)$	$P(S = 1\|C)$
1	0.9	0.1

$$\sum_{S=0}^{1} \sum_{R=0}^{1} \underbrace{P(W = 1|S, R)} \overbrace{P(S|C = 1)} \underbrace{P(R|C = 1)}$$

S	R	$P(W = 1\|S, R)$
0	0	0.0
1	0	0.9
0	1	0.9
1	1	0.99

C	$P(R = 0\|C)$	$P(R = 1\|C)$
1	0.2	0.8

$$= 0 + (0.9 \times 0.1 \times 0.2) + (0.9 \times 0.9 \times 0.8) + (0.99 \times 0.1 \times 0.8)$$
$$= 0.018 + 0.648 + 0.0792 = 0.7452$$

Exact Inference in General

Exact inference: Bayesian inference algorithms that calculate the exact value of probability $P(X|E)$.

Inference by Enumeration:

- We infer a posterior probability by marginalization of the joint distribution.
- That is, we compute the exact value of probability $P(X|E)$ using the equation (1).

Computational Complexity

- The number of terms in the sum is *exponential* in the number of non-evidence random variables: The complexity is $O(nm^n)$, where:
  - n is the number of non-evidence variables.
  - m is the number of values each variable can take.
- Example: In the "Wet Grass" problem, if we do not cancel $P(C = 1)$, the full calculation requires summing 4 terms, each involving 4 multiplications.

Exact inference beyond enumeration

More advanced methods for exact inference aim to speed up computations. We don't cover these, but if you are interested, you can read about these methods in the following textbook:

Daphne Koller & Nir Friedman: Probabilistic Graphical Models
(find the pdf yourself)

- **Variable Elimination**: Chapters 9.2–9.3
  Eliminates variables systematically to reduce computations.
- **Clustering Algorithm**: Chapter 10.1.1
  Groups variables into clusters to simplify computations.
- **Sum-Product Algorithm (Belief Propagation)**: Ch. 10.2 – 10.3
  Efficiently computes marginal probabilities in tree-structured networks.

Summing up Exact Inference

We learned:

- The problem of inference in Bayesian Networks
- A method of exact inference, which uses exhaustive enumeration to marginalise the non-evidence random variables.

We carried out all details of exact inference on a simple problem, the "Wet grass" problem.

Next week: Consolidation week!

- we will look at more examples
- we will solve more exercises
- optionally we will also learn about a useful method of approximate inference in Bayes nets